Per-target descent cost d of compact-orbit Koblitz IC: profile, reductions and the n = 73/83 ranking (2026-10-30)

Verdict.

- **Measured:** the cheapest credible reductions cut the counted S0·d by **2.5x at n = 41, 2.4x at n = 53 and 3.75x at n = 61** against the cold-tuned KIC_NORM recipe, in the same producer and the same unit. They are:
  - a larger table, which trades S0 for d (1.2x to 1.9x);
  - the producer's existing pair-symmetric build, which halves the build and leaves d unchanged (2x).
- **Count projection, not implemented:** batching the S3 inversions on the n ≤ 63 path, as the n ≥ 71 path already does, gives about 2x more. That projection puts all three sizes at S0·d/N' ≈ 1.0.
- **This is a constant factor, not a change in scaling.** At the best recipes S0·d/N' is flat at 1.9 to 2.4 across n = 41, 53, 61 and 83. The reason is that the descent is a table collision:
  - the build costs B S3 solves;
  - each target costs about r/(8Bn) root calls;
  - so S0·d ≥ c^2N'/4, where c is the price of one root call in GAE.

  No retuning in this family moves that bound.
- **Many-target crossover against batch rho:** still no k*. The best measured G is 1.94 / 2.05 / 2.21 at n = 41 / 53 / 61 and 2.02 at n = 83 (w = 0). The projection gives about 1.4. A window needs a root call cheaper than **1.41 GAE** at w = 0. Today's field-operation counts put it at 2.1 GAE (batched) and 2.9–3.3 GAE (unbatched). The n = 83 clock calibration gives about 4.5 to 4.9 rho steps per root call.
- **Section 5a ranking (variant B: w priced on both arms):** rho is ahead at every weight at n = 41, 53, 61, 73 and 83.
  - n = 73 is now counted, at the two recipes run: G = 49.9 (K20 norm32) and 11.0 (K40 norm32) at w = 0, and 48.6 and 10.6 at B, w = 1. Both are dominated by the rank stage and far from the floor.
  - n = 83 (recorded K3300): G = 2.02 at w = 0 and 1.97 at B, w = 1. K600: 16.8 and 16.3.
  - The IC-favourable variant A (w on rho only) is knife-edge at w = 1. Best rows are G = 0.97 (n = 41), 1.02 to 1.10 (n = 53 to 83).
  - With a root call priced at 1 GAE, every best row is below 1.
  - So the ranking depends on the price of a root call. The field-operation count and the clock calibration both put that price above the threshold.
- **Single target (k = 1):** the pair-symmetric build lowers the cold cost S = S0 + d of the tuned KIC_NORM recipe by 1.76x / 1.54x / 1.69x at n = 41 / 53 / 61 (1.45x / 1.33x / 1.43x at B, w = 1).
  - This is a recipe change: it switches on an existing flag (`KIC_PAIR_SYMMETRY=1`, already used by the recorded n = 83 K3300 run) that the KIC_NORM tuning in `research/ic_improve_20261023` left off. It is not a new mechanism.
  - It is a constant factor measured at three sizes. It does not change scaling.

Class:

- **improvement** (constant factor, single-target and multi-target) against the repo's tuned KIC_NORM producer, in counted units;
- **accounting** for the n = 73/83 band ranking.

It is not compared against published IC here. Owner request: IC research-program follow-up B (cut d, rank n = 73/83).

1. What the producer does per target (the d profile)

`examples/koblitz_orbit_dlp_fast_online.rs` (`compact_orbit_online_pair_symmetry_v1`) decomposes a target Q as a sum of four factor-base points by meet in the middle:

- **Index (setup).** For every pair of orbit representatives (P_i, P_j) and relative Frobenius shift, one S3 solve gives the two x-coordinates of P_i ± P_j. They are stored by canonical normal-basis rotation, which quotients out Frobenius.
  - Build: B = W²n solves with W = K·(number of Z[τ] multipliers).
  - With pair symmetry: W(W+1)n/2 solves. Mate states share their canonical roots, so the table is identical.
- **Target.** For each state U and each of the n shifts:
  - one S3 solve S3(x_U, x_Q, v) gives the two x-coordinates of Q ∓ U;
  - each one is probed (rotation canonicalisation plus hash lookup).
  - A hit is lifted and checked (4 points, ≤ 3 additions per sign choice), then the logs are combined.
- **Rank stage.** The same search on [a]G − R_j until the K column logs are fixed: about K + 1 searches.

Table 1 confirms d is almost all the probe scan, in counts and on the clock:

- the PDP share of the per-target interval is 94–100% (56–84% only at n = 41 K48/K96, where a target needs only 78–319 root calls and the fixed recovery check, about 0.02 ms, shows);
- every target needs exactly one relation check;
- the log combination and the recovery check are microseconds.

What does not apply:

- **Early-abort smoothness:** n/a. There is no smoothness test; a probe is an exact table lookup.
- **Relation-database reuse for the individual log:** already the design. One relation per target is resolved by the column logs from the rank stage.

The bound (derived, then checked against counts).

- The table holds $E \approx 2B/(1 + \mathrm{sym})$ canonical roots. Each covers n x-coordinates, so $E{\cdot}n$ x-values out of the r/2 x-coordinates of the subgroup.
- A probe hits with probability $E{\cdot}n/(r/2)$. Each root call makes two probes, so a target costs about $r/(4En)$ root calls.
- With a plain build, $E \approx B$: build × d_rc $\approx r/(4n) = N'/2$. With a symmetric build: $N'/4$.
- Measured build × d_rc / N' (last column of Table 2):
  - plain build: 0.43–0.49 at the largest tables of n = 41, 53, 61;
  - symmetric build: 0.22–0.25;
  - small tables at n = 61 run higher (0.66–0.84), and n = 83 K3300 (a = 1, symmetric) is 0.35.

So $S0{\cdot}d \geq c^2N'/4$ + rank work. Since $\mathrm{rho1}^2 \approx \pi N'/2$, $\sigma\delta \geq c^2/(2\pi)$. The window condition $\sigma\delta < 0.318$ then needs $c < 1.41$ GAE at w = 0. Under variant B at w = 1 it needs $c + 2w < 1.41(1 + w)$, i.e. $c < 0.83$.

2. Pricing a root call

These counts are from the producer's source:

operation	field multiplications	GAE (1 GAE = batched affine addition = 5 M)
S3 root call, u64 path (n ≤ 63), unbatched	8.5 M + Itoh–Tsujii inversion (6/7/8 M at n = 41/53/61)	2.90 / 3.10 / 3.30
S3 root call, wide path (n ≥ 71), inversion batched over 2n	2 + 3 + 5.5 M + I/(2n)	2.11
relation check (3 unbatched additions)	3(I + 2 M)	4.8–6.0
probe (n-step rotation canonicalisation + hash lookup); table insert	none (word operations, linear maps)	w ∈ {0, 0.1, 1}

Calibration (§5a allows it as an extra point, not a replacement): the n = 83 records give clock time per counted operation, each pair measured on one runner.

- IC K3300 spends 0.94 µs and 1.24 µs of PDP time per root call, including its two probes, in the two runs.
- Plain rho spends 0.19 µs and 0.275 µs per addition, including its canonicalisation.
- That is a ratio of 4.9 and 4.5.

Both the count and the clock place a root call well above the 1.41 GAE threshold.

3. Results

Table 1: where d goes (per target, mean over the run's targets)

n	recipe	targets	root calls	probes (mean / median / max)	relation checks	skipped mate states	PDP ms	relation-check ms	log combination ms	recovery check ms	PDP share of target ms
41	K125 norm0	32	3305	6610 / 3412 / 24349	1.00	0.0	1.747	0.0082	0.00061	0.0206	98.3%
41	K125 norm0 sym	32	3254	6508 / 3412 / 24349	1.00	0.6	1.141	0.0072	0.00148	0.0207	97.5%
41	K12 norm16	32	5556	11112 / 7072 / 42351	1.00	0.0	3.244	0.0151	0.00050	0.0212	98.9%
41	K12 norm16 sym	32	5556	11112 / 7072 / 42351	1.00	0.0	4.965	0.0089	0.00062	0.0217	99.4%
41	K24 norm16	32	1390	2779 / 2200 / 6681	1.00	0.0	0.844	0.0237	0.00063	0.0212	94.9%

n	recipe	targets	root calls	probes (mean / median / max)	relation checks	skipped mate states	PDP ms	relation-check ms	log combination ms	recovery check ms	PDP share of target ms
41	K24 norm16 sym	32	1390	2779 / 2200 / 6681	1.00	0.0	0.977	0.0092	0.00067	0.0275	96.3%
41	K48 norm16	32	319	638 / 370 / 2244	1.00	0.0	0.139	0.0071	0.00026	0.0198	83.6%
41	K96 norm16	32	78	155 / 92 / 1371	1.00	0.0	0.037	0.0080	0.00028	0.0201	56.5%
41	K96 norm16 sym	32	78	155 / 92 / 1371	1.00	0.0	0.074	0.0113	0.00071	0.0270	65.4%
53	K20 norm16	32	34373	68745 / 51906 / 224106	1.00	0.0	30.156	0.0144	0.00085	0.0424	99.8%
53	K20 norm16 sym	32	34373	68745 / 51906 / 224106	1.00	0.0	17.654	0.0150	0.00095	0.0353	99.7%
53	K280 norm0	32	22980	45960 / 29661 / 179464	1.00	0.0	9.141	0.0113	0.00086	0.0319	99.5%
53	K280 norm0 sym	32	22894	45787 / 29661 / 179464	1.00	0.8	10.901	0.0154	0.00157	0.0328	99.5%
53	K40 norm16	32	10841	21681 / 16684 / 74439	1.00	0.0	9.969	0.0119	0.00085	0.0354	99.5%
53	K40 norm16 sym	32	10841	21681 / 16684 / 74439	1.00	0.0	9.898	0.0291	0.00091	0.0505	99.2%
53	K80 norm16	32	2518	5036 / 2710 / 21850	1.00	0.0	1.119	0.0108	0.00043	0.0317	96.3%
53	K80 norm16 sym	32	2518	5036 / 2710 / 21850	1.00	0.0	2.764	0.0170	0.00072	0.0333	98.2%
61	K112 norm16	32	7618	15235 / 9282 / 53068	1.00	0.0	4.173	0.0214	0.00123	0.0432	98.4%
61	K112 norm16 sym	32	7618	15235 / 9282 / 53068	1.00	0.0	5.952	0.0267	0.00382	0.0509	98.7%
61	K18 norm32	32	117222	234443 / 169522 / 1544663	1.00	0.0	54.440	0.0175	0.00086	0.0435	99.9%
61	K18 norm32 sym	32	117222	234443 / 169522 / 1544663	1.00	0.0	57.830	0.0145	0.00093	0.0408	99.9%
61	K28 norm16	32	233374	466747 / 330234 / 1538843	1.00	0.0	116.289	0.0333	0.00410	0.1118	99.9%
61	K28 norm16 sym	32	232916	465832 / 330234 / 1531523	1.00	3.8	284.965	0.4395	0.02347	0.0551	99.8%
61	K500 norm0	32	42826	85651 / 56056 / 419144	1.00	0.0	18.909	0.0141	0.00089	0.0404	99.7%

n	recipe	targets	root calls	probes (mean / median / max)	relation checks	skipped mate states	PDP ms	relation-check ms	log combination ms	recovery check ms	PDP share of target ms
61	K500 norm0 sym	32	42826	85651 / 56056 / 419144	1.00	0.0	21.684	0.0142	0.00131	0.0411	99.7%
61	K56 norm16	32	46126	92251 / 62030 / 345423	1.00	0.0	56.626	0.0301	0.00139	0.0489	99.9%
61	K56 norm16 sym	32	46126	92251 / 62030 / 345423	1.00	0.0	31.159	0.0242	0.00139	0.0812	99.7%
73	K20 norm32	8	120516174	241032265 / 87566307 / 846313503	1.00	0.0	77956.824	0.0562	0.00658	0.0974	100.0%
73	K40 norm32	8	10822542	21644946 / 8567750 / 78371179	1.00	0.0	7925.265	0.0466	0.00759	0.0959	100.0%
83	K600 (recorded, a=1)	16	1989790	3979401 / 1125816 / 18098258	1.00	0.0	2024.687	0.0302	0.00072	0.1254	100.0%
83	K3300 (recorded, a=1, sym)	16	39742	79485 / 52294 / 267565	1.00	0.0	37.181	0.0069	0.00059	0.0892	99.7%

Table 2: counted S0 and d, every run (GAE; root call priced in field multiplications; w = 0 unless named)

n	recipe	root call [GAE]	build root calls	rank root calls	S0	d	S0*d*sigma*delta N'	G, w = 0 [95%]	G, B w = 0.1	G, B w = 1	G, A w = 1	G, root call = 1 GAE (IC-favourable)	root-call price for G = 1 at A w = 1	build x d root calls / N'
41	K125 norm0	2.90	6.406e5	6.614e5	3.776e6	9.590e3	5.40	3.44	3.29 [2.74, 3.77]	3.17	2.63	1.64	1.13	1.76	0.316
41	K125 norm0 sym	2.90	3.204e5	6.566e5	2.834e6	9.442e3	3.99	2.54	2.83 [2.33, 3.31]	2.75	2.39	1.41	0.98	2.05	0.156
41	K12 norm16	2.90	5.904e5	8.759e4	1.966e6	1.612e4	4.73	3.01	3.08 [2.58, 3.57]	2.95	2.35	1.54	1.06	1.88	0.489
41	K12 norm16 sym	2.90	2.953e5	8.759e4	1.110e6	1.612e4	2.67	1.70	2.31 [1.87, 2.70]	2.25	1.95	1.16	0.80	2.51	0.245
41	K24 norm16	2.90	2.362e6	5.096e4	6.997e6	4.036e3	4.21	2.68	2.90 [2.59, 3.21]	2.78	2.19	1.45	1.00	2.00	0.490
41	K24 norm16 sym	2.90	1.181e6	5.096e4	3.573e6	4.036e3	2.15	1.37	2.07 [1.86, 2.31]	2.01	1.75	1.04	0.72	2.80	0.245
41	K48 norm16	2.90	9.446e6	2.051e4	2.745e7	9.303e2	3.81	2.43	2.76 [2.27, 3.19]	2.64	2.08	1.38	0.96	2.10	0.450
41	K96 norm16	2.90	3.779e7	8.732e3	1.096e8	2.307e2	3.77	2.40	2.75 [2.14, 3.34]	2.62	2.06	1.37	0.97	2.10	0.439
41	K96 norm16 sym	2.90	1.889e7	8.732e3	5.482e7	2.307e2	1.89	1.20	1.94 [1.52, 2.44]	1.89	1.63	0.97	0.68	2.99	0.220

n	recipe	root call [GAE]	build root calls	rank root calls	S0	d	S0*sigma*delta N'		G, w = 0 [95%]	G, B w = 0.1	G, B w = 1	G, A w = 1	G, root call = 1 GAE (IC-favourable)	root-call price for G = 1 at A w = 1	build x d root calls / N'
53	K20 norm16	3.10	2.120e6	8.784e5	9.295e6	1.066e5	4.99	3.18	3.16 [2.65, 3.62]	3.03	2.41	1.58	1.02	1.96	0.367
53	K20 norm16 sym	3.10	1.060e6	8.784e5	6.009e6	1.066e5	3.23	2.05	2.55 [2.13, 2.89]	2.46	2.09	1.27	0.82	2.44	0.184
53	K280 norm0	3.10	4.155e6	6.741e6	3.378e7	7.124e4	12.12	7.72	4.93 [4.01, 5.74]	4.74	3.90	2.46	1.59	1.26	0.481
53	K280 norm0 sym	3.10	2.078e6	6.738e6	2.733e7	7.098e4	9.77	6.22	4.42 [3.60, 5.21]	4.28	3.64	2.21	1.43	1.40	0.240
53	K40 norm16	3.10	8.480e6	6.323e5	2.825e7	3.361e4	4.78	3.04	3.09 [2.59, 3.56]	2.95	2.30	1.55	1.00	2.00	0.463
53	K40 norm16 sym	3.10	4.240e6	6.268e5	1.509e7	3.361e4	2.55	1.63	2.26 [1.81, 2.61]	2.19	1.86	1.13	0.73	2.74	0.232
53	K80 norm16	3.10	3.392e7	2.183e5	1.058e8	7.813e3	4.16	2.65	2.89 [2.25, 3.43]	2.75	2.13	1.44	0.93	2.15	0.430
53	K80 norm16 sym	3.10	1.696e7	2.183e5	5.325e7	7.813e3	2.10	1.33	2.05 [1.62, 2.42]	1.98	1.68	1.02	0.66	3.03	0.215
61	K112 norm16	3.30	7.652e7	9.840e5	2.558e8	2.515e4	4.82	3.07	3.10 [2.60, 3.58]	2.95	2.25	1.55	0.94	2.13	0.437
61	K112 norm16 sym	3.30	3.826e7	9.840e5	1.295e8	2.515e4	2.44	1.55	2.21 [1.87, 2.58]	2.13	1.77	1.10	0.67	2.99	0.218
61	K18 norm32	3.30	7.906e6	1.656e6	3.155e7	3.868e5	9.14	5.82	4.28 [3.32, 5.25]	4.08	3.16	2.14	1.30	1.54	0.694
61	K18 norm32 sym	3.30	3.953e6	1.649e6	1.849e7	3.868e5	5.36	3.41	3.28 [2.65, 3.98]	3.16	2.63	1.64	0.99	2.01	0.347
61	K28 norm16	3.30	4.782e6	4.958e6	3.214e7	7.701e5	18.54	11.80	6.11 [5.01, 7.06]	5.85	4.67	3.05	1.85	1.08	0.836
61	K28 norm16 sym	3.30	2.391e6	4.954e6	2.424e7	7.686e5	13.95	8.88	5.30 [4.46, 6.08]	5.11	4.26	2.65	1.61	1.24	0.417
61	K500 norm0	3.30	1.525e7	2.204e7	1.231e8	1.413e5	13.03	8.29	5.11 [4.35, 5.95]	4.89	3.94	2.55	1.55	1.29	0.489
61	K500 norm0 sym	3.30	7.625e6	2.202e7	9.783e7	1.413e5	10.36	6.59	4.55 [3.81, 5.29]	4.39	3.66	2.28	1.38	1.45	0.245
61	K56 norm16	3.30	1.913e7	2.234e6	7.050e7	1.522e5	8.04	5.12	4.01 [3.34, 4.57]	3.82	2.94	2.01	1.22	1.65	0.661
61	K56 norm16 sym	3.30	9.565e6	2.234e6	3.894e7	1.522e5	4.44	2.83	2.98 [2.53, 3.42]	2.87	2.39	1.49	0.90	2.21	0.330
73	K20 norm32	2.11	1.168e7	1.328e9	2.826e9	2.542e8	1219.30	776.23	49.94 [25.34, 69.88]	49.69	48.59	24.97	23.67	0.08	2.389

n	recipe	root call [GAE]	build root calls	rank root calls	S0	d	S0*d/N'	sigma*delta	G, w = 0 [95%]	G, B w = 0.1	G, B w = 1	G, A w = 1	G, root call = 1 GAE (IC-favourable)	root-call price for G = 1 at A w = 1	build x d root calls / N'
73	K40 norm32	2.11	4.672e7	6.879e8	1.550e9	2.283e7	60.05	38.23	10.98 [6.39, 14.78]	10.91	10.61	5.49	5.20	0.38	0.858
83	K600 (recorded, a=1)	2.11	2.988e7	7.249e8	1.592e9	4.198e6	129.48	89.92	16.81 [8.78, 22.69]	16.72	16.30	8.41	7.97	0.25	1.152
83	K3300 (recorded, a=1, sym)	2.11	4.519e8	9.543e7	1.155e9	8.385e4	1.88	1.30	2.02 [1.64, 2.46]	2.01	1.97	1.01	0.96	2.09	0.348

Table 3: reductions of S0*d against the cold-tuned KIC_NORM recipe (same producer, same unit, w = 0)

n	baseline S0*d/N'	best retuned K, plain build	pair symmetry, same K	best retuned K + pair symmetry (measured)	with batched root calls (count projection, not run)	G: baseline -> best measured -> projected
41	4.73	K96 norm16: 3.77 (1.25x)	2.67 (1.77x)	K96 norm16 sym: 1.89 (2.51x)	1.00 (4.71x; root call 2.90 -> 2.11 GAE)	3.08 -> 1.94 -> 1.42
53	4.99	K80 norm16: 4.16 (1.20x)	3.23 (1.55x)	K80 norm16 sym: 2.10 (2.38x)	0.97 (5.12x; root call 3.10 -> 2.11 GAE)	3.16 -> 2.05 -> 1.40
61	9.14	K112 norm16: 4.82 (1.90x)	5.36 (1.71x)	K112 norm16 sym: 2.44 (3.75x)	1.00 (9.14x; root call 3.30 -> 2.11 GAE)	4.28 -> 2.21 -> 1.41
73	1219.30	K40 norm32: 60.05 (20.30x)	not run	not run	n/a (wide path already batched)	49.94 -> 10.98

Table 4: single-target (k = 1) cold cost S = S0 + d, GAE

n	baseline recipe	baseline S, w = 0	best measured recipe	best S, w = 0	gain, w = 0	gain, B w = 1
41	K12 norm16	1.982e6	K12 norm16 sym	1.126e6	1.76x	1.45x
53	K20 norm16	9.402e6	K20 norm16 sym	6.116e6	1.54x	1.33x
61	K18 norm32	3.194e7	K18 norm32 sym	1.887e7	1.69x	1.43x
73	K20 norm32	3.080e9	K40 norm32	1.573e9	1.96x	1.99x
83	K600 (recorded, a=1)	1.597e9	K3300 (recorded, a=1, sym)	1.155e9	1.38x	1.37x

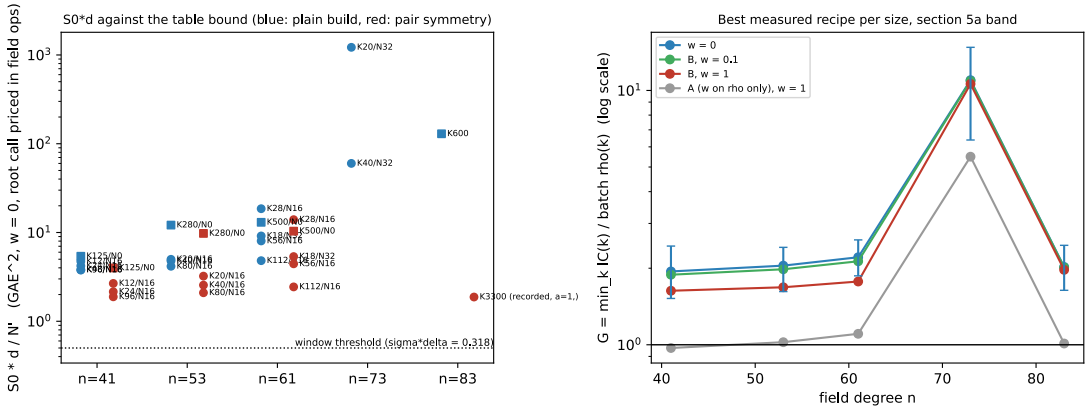


Figure 1: S0*d against the table bound, and G across the section 5a band

Figure. Left panel: S0·d/N' for every run (GAE², w = 0). Blue points are plain builds and red points are pair-symmetric builds; squares are plain (non-Z[τ]) bases. The dotted line is the window threshold σδ = 0.318, which is S0·d/N' = 0.5. Right panel: G for the best measured recipe at each size across the §5a band. Error bars are 95% bootstrap intervals over targets (400 draws). Sample sizes: 32 targets per run at n ≤ 61, 8 at n = 73, 16 at n = 83.

4. Skeptical reading and limits

1. **What the result rests on.**
   - **Counts.** Root calls, probes, inserts and relation checks are emitted by the producer and deterministic. They are identical between the plain and symmetric runs at every K. `skipped_symmetric_states` is 0 to 4 per target, so d is unchanged by symmetry.
   - **Prices.** The GAE prices come from a multiplication count in the source, not from a measured primitive benchmark. The clock calibration in §2 is the only empirical check on them.
2. **Rho reference.**
   - At n ≤ 73, rho1 is the floor sqrt(πN'/2), with one canonicalisation per addition. The repo's measured counted/floor ratio is 1.02 to 1.08, so using the floor favours rho by up to 8% in G ($\sqrt{1.08} \approx 4\%$).
   - At n = 83 it is the mean of the 16 recorded plain-rho walks (8.62e6 additions against a floor of 9.0e6).
3. **Batched root calls are a count projection.** The n ≤ 63 path was not changed or rerun. The n ≥ 71 path already batches, and its count per call is the projected one.
4. **n = 73 did not reach the floor.**
   - The two runs are K20 norm32 (W = 400, the untuned recipe of `research/ic_improve_20261023`) and K40 norm32 (W = 800), each with 8 targets.
   - **`KIC_PAIR_SYMMETRY=1` was passed to the K40 run, but the wide path ignores it in this binary.** Its summary has no `pair_symmetry` field and `regular_states` = W^2n. The row is labelled as a plain build, and the files were renamed `_s0_`.
   - Both rows are dominated by the rank stage (rank root calls 1.3e9 and 6.9e8, against builds of 1.2e7 and 4.7e7), so G is 50 and 11.
   - The descent is also less efficient there than the coverage bound:
     - build × d_rc / N' is 2.39 at K20 and 0.86 at K40, against 0.5 for a plain build;
     - at n = 61 the small-K Z[τ] recipes show the same thing more mildly (0.66–0.84);
     - it improves with more seeds per multiplier. The cause was not isolated. Additive structure among the Z[τ] images of few seeds is a guess, not a finding.
   - Peak RSS was 2.7 GB (W = 400) and 10.9 GB (W = 800). A table near the floor at n = 73 (W of a few thousand) was not attempted, because host memory is unknown and about 10 GiB of disk is free.
   - The n = 73 rows give the counted §5a ranking that part (0) asked for, not a tuned n = 73 point.
   - The single-target gain at n = 73 (1.96x) is a retune of an untuned recipe.
5. **n = 83 is a = 1 and recorded.**
   - Counts are from `research/20261016-n83-precomputed-rho-control/records/run-38855595549`. Run `-38878565324` has identical counts.
   - The K3300 summary has no rank root-call total. It is taken as probes/2, which holds exactly in K600, where both totals exist: S3(x_U, x_Q) always has its two roots when U and Q are points.
6. **Selection.** The "best" recipe at each size is the minimum over 4 to 7 measured recipes, one seed, and 32 targets. The Table 2 intervals (about ±20%) are the guide.
7. **Not compared with published IC.** FGHR, GGMP and Petit-type summation-polynomial descents have no table counterpart at these sizes. The baseline here is the repo's tuned KIC_NORM producer (`research/ic_improve_20261023`).
8. **No benchmarks row.** `benchmarks/README.md` admits rows only through the validation gate (a replay certificate, a validator report, and a matched rho named by `VALIDATION.json`). None of that was produced here, so the row is a follow-up and not part of this commit.

5. Where to go next

- **The constant that decides the crossover is the root-call price c.** The window needs c < 1.41 GAE at w = 0, and c < 0.83 under variant B at w = 1. A target-side S3 solve with x_Q fixed shares half its products across a target and needs about 10 M. Getting below about 7 M per call (1.41 GAE) would need a different solve, for example sharing the x_Q-dependent products across the n shifts of one state. That was not tried, and it is not known to be possible.
- **Batch the n ≤ 63 path** (about 2x in S0·d, by count). It is a small change; the wide path's `RootBatch128` is the template.
- **Leaving the table family.** Anything that beats S0·d ≈ c^2N'/4 has to recognise decomposable points without enumerating them. That is a summation-polynomial or Gröbner/SAT descent per trial, whose per-trial cost at these n is many orders above c (`ic descent` tables).

Files

- `descost/`: native Rust (std + `serde_json`). It reads the run outputs and the n = 83 records and writes `results.json` and `tables.md`. The tables in §3 are pasted from `tables.md` unchanged. Build with `cargo build --release --offline` in `descost/`, then run `descost runs <crypto>/research/20261016-n83-precomputed-rho-control/records/run-38855595549 .`
- `run.sh`, `all.sh`, `extra.sh`: the run commands. `run.sh <n> <K> <norm> <targets> <sym>` runs `KIC_NORM=<norm> KIC_PAIR_SYMMETRY=<sym> koblitz_orbit_dlp_fast_online construct:<n>:0:<K> <targets> 1 <out>`.
  - Binary: `target/release/examples/koblitz_orbit_dlp_fast_online`, built 2026-10-26 from source at or after `2385f77e46`, sha256 `ce8f8d23f6aa9abcc470b7f24fcf79e13e3125d2822615201fe0ec20fc1a8da9`.
  - Host: shared Apple-silicon Mac, load average about 19, single thread, no isolation. Wall times are practicality notes only.
- `runs/`: one `<tag>.stdout` (summary line), `<tag>.jsonl` (per-target records), `<tag>.targets` and `<tag>.stderr` per run. `targets/`: seeded known-answer scalars (seed 20261030 + n).
- `plot.py` (layout only, reads `results.json`), `s0d_and_g.svg`/`.png`, `report.pdf` (this README rendered).

- Subgroup orders r (largest prime factor of #E(F_2^n)): 549756390943, 21044858204113, 162888033982417, 86020738150056119 (a = 0, n = 41/53/61/73), and 8569786107849059 (a = 1, n = 83).